Effective Context Utilization, Conversation Length, and Factual Reliability in LLM Assistants

Structured narrative evidence review | Research ID: llm-context-memory-reliability-2026-11

Research execution based on approved brief BR-001

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Executive synthesis

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Executive synthesis

Research question (Q0). Under what conditions do context length, information placement, and multi-turn interaction affect factual reliability, and how far can those effects be distinguished from cross-session memory failures?

Bottom line. There is strong controlled evidence that *effective use* of an available context is task-, position-, content-, and model-dependent; supported input length is not a guarantee of accurate use. There is also controlled evidence for failures specifically induced by distributing requirements across conversational turns—even at ordinary, non-saturating context lengths. Persistent memory is a distinct, multi-stage system: an error observed in a later session does not identify whether a fact was never stored, became stale, was not retrieved, was omitted from the generation prompt, was misread, or

was used inappropriately. No defensible general claim follows that reliability must decrease monotonically with conversation length, nor that every conversation error is a memory failure.

Five consequential observations. First, in Liu et al.'s controlled 20-document question-answering experiment, GPT-3.5-Turbo answered 75.8% correctly with the answer-bearing passage first, 53.8% when it was tenth, and 63.2% when it was last, holding the document set and question fixed as positions varied: a 22.0-percentage-point best-to-worst contrast [R1]. Second, harder tests can contradict near-perfect single-needle results: RULER probes tracking and aggregation; NoLiMa removes literal overlap; Du et al. identify answer degradation after verifying evidence retrieval, including reduced-distraction controls [R3,R6,R8]. Third, Laban et al.'s 15-model, six-task simulated study found a mean **39% relative performance decrease** from fully specified single-turn tasks to sharded multi-turn versions. Concatenating the same shards into one prompt retained **95.1%** of full-prompt performance, providing a useful, though not perfect, control for information preservation and paraphrase [R11]. Fourth, long-term-memory studies separate access from use: LongMemEval reports retrieval and downstream reading separately, while 2026 work examines updates, revoked facts, scope restrictions, and natural conversational integration [R16,R18-R21]. Fifth, apparent counter-examples matter: certain models stay strong on RULER out to 128K, gold conversational context can initially *improve* accuracy as turns accumulate, and a four-month deployment found large differences in direct-QA scores without corresponding changes in satisfaction [R3,R14,R21].

Causal conclusion. Position manipulation provides direct evidence that position *can cause* output changes in the tested prompts. Sharding and gold/self-conditioned turn comparisons identify effects of particular interaction protocols, but do not estimate an invariant pure "per-turn penalty." Memory-stage interventions can narrow the fault location, but benchmarks without write traces, retrieval traces, and exact exposed prompts cannot locate a failure reliably. These findings are predominantly *behavioral*. Mechanisms such as attention saturation, positional encoding, early-answer anchoring, and hidden product memory management remain hypotheses unless directly manipulated or observed.

1. Scope, outcome definitions, and evidence standard

This is a **structured narrative review**, not a systematic review or a new experimental reproduction. It covers text-based assistants, open-weight and proprietary models, studies and available documents **through 3 November 2026**, and commercial products as illustrative systems. The objective is attribution, not a product leaderboard. Inclusion, exclusion, model-version, and inspection constraints are recorded in the separate analysis companion. The approved research boundaries control this report; findings from public sources are research evidence, not content inherited from that brief.

A context *window* is a processing limit on input/token handling; *effective utilization* is the task-conditional ability to recover, reconcile, and use evidence within that window. A conversational *turn* is a structured exchange with role tags, ordering, potentially assistant-generated assumptions, and incremental disclosure; its effects are not equivalent to adding the same number of tokens in one message. *Cross-session memory* entails selecting information from a prior interaction to preserve and later expose to the model, perhaps through summaries, search, or structured memory stores. An assistant may possess all relevant information in its current prompt and still produce a wrong answer; conversely, a strong model cannot use material that a memory/retrieval layer never exposes.

“Factual reliability” is not one interchangeable benchmark number. Studies respectively measure exact answer correctness, token/string matching, task execution, grounded claim precision, calibrated abstention, time-valid correctness, policy-governed use, or the variance of answer quality across runs. We therefore report denominators, scoring rules, and comparison type, and avoid averaging unrelated published percentages. FActScore illustrates why the unit of assessment matters: a long output can contain both correct and unsupported atomic facts even if its overall answer seems plausible [R22]. A user’s satisfaction is a separate deployment outcome, not a substitute for verifiable correctness [R21].

Evidence weights are qualitative: **higher** for primary studies with controlled perturbations and inspectable protocols; **moderate** for task-specific, synthetic, or limited-dataset experiments with useful controls; **bounded** for small commercial pilots, vendor-run tests, or longitudinal observations; **anecdotal** for single reports without controlled traces. “Higher” refers to internal identification for the tested task, not unrestricted external validity.

2. Q1 — How effective context varies with length, position, distraction, and conflict

2.1 Position effects: one fixed evidence set, different order

Liu et al. manipulated the slot of the single answer-containing passage among distractor passages in multi-document NaturalQuestions-based QA (10, 20, and 30 documents). They also tested synthetic key-value extraction, and compared several 2023-era model variants [R1]. This holds the question, relevant source, and candidate set substantially constant while manipulating the salient evidence position. With 20 passages, GPT-3.5-Turbo’s five tested position accuracies were **75.8, 57.2, 53.8, 55.4, and 63.2%**, at positions 1, 5, 10, 15, and 20 respectively. A closed-book baseline scored 56.1% in their comparison; the worst middle slot was below it. For another model, task, or prompt, the curve need not be U-shaped. The study included distractor-order and ambiguity checks, but originally tested older models and a selected QA protocol; it did not examine cross-session storage.

The relevant estimand is a *within-item placement contrast*, not simply accuracy at 4K tokens. Write $Y_i(p)$ for correctness on item i when gold evidence is at position p, and define $\Delta(p,q) = (1/n)\,\Sigma_i\,[Y_i(p) - Y_i(q)]$. With paired prompt instances and controlled decoding, the contrast establishes that ordering can change output quality. It does **not** by itself establish whether retrieval failed, whether competing facts gained salience, whether the answer generator ignored recognized evidence, or which attention component caused the effect.

2.2 Length and task difficulty: window size is not effective reading

RULER varies input length (4K–128K tokens), includes 13 synthetic tasks in retrieval, multi-hop variable tracking, aggregation, and QA, and originally evaluates 17 long-context models using approximately 500 instances per task/length [R3]. Its "effective context length" is *operational*: maximum length passing a threshold defined by Llama-2-7B's 4K score (85.6%), not a universal physical limit. In the inspected study table, **Gemini-1.5-Pro scored 96.7% at 4K and 94.4% at 128K**, whereas **Llama-3.1-70B scored 96.5% and 66.6%** across the same task aggregate. This is a critical negative case against a blanket claim that all models degrade comparably. The authors explicitly note that the tasks are synthetic and that their published aggregate does not independently control gold-evidence position; the live code repository can also accumulate later model results that should not be mistaken for results of the original paper.

LongBench covers 21 bilingual task datasets; LongBench v2 deliberately increases multi-document and reasoning difficulty using **503 multiple-choice questions**, spanning contexts from 8K to 2M **words** (not tokens) [R4,R7]. These improve breadth and realism but alter question difficulty with domain and source length, often preventing clean causal estimates of context length alone. BABILong varies buried facts and reasoning chains in long natural text; RAG and recurrent-memory approaches succeed on different portions of its tasks, emphasizing that successful retrieval of one fact is not proof of successful distributed reasoning [R5].

NoLiMa specifically defeats shortcut lexical matching between the question and "needle." Across **13 models** claiming at least 128K context, **11 fell below half their short-context baseline at 32K**; the study reports **GPT-4o falling from 99.3% to 69.7%** [R6]. This result is stronger evidence of a semantic-location bottleneck than vanilla exact-string NIAH, but the benchmark still uses constructed question–needle associations and does not demonstrate a particular neural cause. MGAL's multilingual position- and granularity-aware tests on 8K–128K UN reports add a 2026 boundary condition: word-level extraction can be strong while sentence/paragraph/document-level, source-consistent interpretation remains weak, particularly under local semantic crowding [R10].

2.3 Can processing length matter *after* successful retrieval?

Du et al. (EMNLP 2025) explicitly separated an **evidence-recitation probe**, scored with strict exact match, from the **answer-generation run**. They varied input length on GSM8K mathematics, MMLU QA, HumanEval coding, and a

constructed variable-summation task across two open- and three closed-source models. On settings where relevant evidence and question could be recited, answer performance still declined substantially (paper-reported range **13.9%–85%**, across heterogeneous task/model comparisons) as context length increased [R8]. Additional controls used low-semantic-load whitespace, masked distractor tokens for supported open models, and placement of evidence immediately before the question. A short-context evidence-restatement intervention improved selected RULER evaluations by up to **4 percentage points** on GPT-4o.

This is unusually informative but should be read precisely. Exact recitation in an *independent retrieval trial* is not a direct observation of the model's internal evidence state during its answer trial. A mask may suppress attention to distractor tokens but does not remove positional indices or change every implementation detail. Some closed models declined to recite evidence, preventing the strict retrieval control in those combinations. The authors report only five models/four tasks, and the mitigation presupposes unusually accurate evidence extraction. The observed residual is thus **not fully explained by inability to recite the evidence**, rather than proof that sheer token count is the only microscopic cause.

2.4 Distractors, contradiction, and competing instructions

Shi et al. created GSM-IC by injecting irrelevant information into otherwise solvable grade-school math questions; controlled added context reduced accuracy in tested settings, and self-consistency or explicit irrelevance instructions mitigated some failures [R2]. Kurfalı and Östling varied repetition, position, surface form, and layout of **three mutually incompatible** answers inside a long document. The tested models often chose one candidate without acknowledging the conflict, with substantial repetition/recency influences; their design intentionally made no candidate intrinsically true, so it measures *selection and uncertainty behavior*, not answer truth [R9]. MultiTurnInstruct tests retrieval, cross-turn tracking, and resolution of entangled requirements in 1.1K constructed dialogues; strong memorization and weak privacy/conflict handling can coexist [R13].

The right diagnostic distinction is between (i) **not finding** relevant evidence, (ii) **finding but misinterpreting** it, (iii) **selecting stale or lower-authority evidence**, and (iv) **following an instruction that should have been treated as untrusted source text**. These should have separate labels and source-of-truth specifications. A single exact-match score cannot distinguish them. The available studies do not imply that putting evidence at the top always fixes conflicts; instructions, chronology, task semantics, and document authority may demand a different decision.

3. Q2 — Do conversation turns create problems beyond tokens alone?

3.1 The most informative comparison: FULL, CONCAT, SHARDED

Laban et al. (ICLR 2026) transformed about **100 items for each of six tasks**—code, databases, actions, data-to-text, mathematics, and summarization—into sequences of requirements ("shards"). They evaluated **15 models**, **10 repeated simulations** per model/item/condition, and **more than 200,000 simulated conversations**, typically within **8K tokens** and always below **20K** for the reported tasks [R11]. Their three principal conditions were:

1. **FULL**: all requirements in one fully specified instruction.
2. **CONCAT**: the same sharded requirements concatenated into one instruction; this is a paraphrase and information-preservation control.
3. **SHARDED**: requirements revealed progressively over dialogue; assistant responses can become part of subsequent history.

Across six differently scored tasks, the authors report average performance **39% lower** for SHARDED versus FULL, versus CONCAT maintaining **95.1%** of FULL performance. On repeated stochastic simulations they define mean performance $\bar{P}$, "aptitude" as the 90th percentile $\mathbf{A = P_{90}(S)}$, and **unreliability** as $\mathbf{U = P_{90}(S) - P_{10}(S)}$. The authors report an average **16% aptitude decrease** and **112% relative increase in their unreliability measure** in SHARDED. That last percentage is **not** a probability of error and **not** 112 percentage points. Note also that response quality may be scored by executable tests, semantic equivalence, BLEU, or a task-specific LLM judge; the combined figure is not a universal factual-answer error rate.

This is strong evidence of a **turn/protocol effect above a simple "model saw the task words" explanation**, since concatenation largely preserves performance. It is **not** a perfectly token-matched, turn-count-only randomized experiment: sharding introduces timing of information, assistant-generated text, opportunities for premature answers, role markers, and potentially greater total tokens. Its most plausible observed mediator is early commitment and difficult recovery, but the causal identification is to the *sharded conversation treatment*, not uniquely to the integer number of turns or any single component of hidden attention.

3.2 Oracle history versus self-conditioned history

ChronoScope (Atri et al., ACL 2026) comprises **1,469,628** deterministically generated Wikidata-based question chains of **2–11 turns**, testing temporally scoped fact carryover, switches, and cross-entity transfer [R14]. It compares **Gold Context** (past assistant answers replaced with correct ones), **Self-Conditioned** (the model sees its own prior answers), and **Questions Only** (no prior dialogue context). Final-turn and strict all-turn accuracy are reported separately.

An especially important counterfinding is that with **Gold Context**, some models' final-answer accuracy *improves* from two through four or five turns before plateauing/reversing, likely benefiting from repeated temporal anchoring. Under **Self-Conditioned** histories, longer chains show much larger deterioration. This rejects "each additional turn inevitably hurts" and supports error propagation as a material mechanism. Gold histories are not proof of perfect inference: the paper reports remaining drift toward present-day answers despite gold previous answers. It does not directly hold all token counts fixed, and Wikidata-derived chains are more regular than spontaneous human dialogue.

MultiChallenge (2025) provides complementary ecological stress tests of instruction-following and cross-turn scope; its leading tested model achieved **41.4%** average accuracy, with all evaluated frontier models below 50% on its four challenge types [R12]. This establishes that challenging conversational behaviors are missed by simpler tests, but **does not provide a clean token-matched estimate of added-turn causality**. The existence of high-scoring specialized systems or domains, and ChronoScope's gold-context gains, prevent generalizing one percentage across all conversations.

3.3 What an actually isolated "turn count" experiment needs

Fix model weights/version, decoding, task, final question, correct evidence, content ordering, total tokens (including assistant turns), and role framing as far as feasible. Randomly distribute the *same* semantic facts among 1, 2, 4, and 8 turns using length-matched padding; independently cross assistant-generated text versus oracle/neutral text. Then estimate $\mathbf{E[Y \mid do(T = t), L = \ell, P = p, D = d, A = a]}$. Include a condition with all requirements disclosed at the start plus empty/neutral later turns to isolate a weaker "extra boundaries" effect. A full factorial design is expensive, and padding/role distribution can themselves influence the model; report both practical protocol effects and narrowly defined controlled contrasts. ChronoScope and Laban provide pieces of such identification, not the full experiment.

4. Q3 — Persistent cross-session memory: which stage failed?

4.1 The operational pipeline and its failure map

A useful decomposition for a persistent assistant is:

Prior sessions → extraction/writing → stored representations and versioning → query-triggered retrieval → eligibility/priority/selection → material actually exposed in the current prompt → interpretation and generation → answer.

LongMemEval formalizes **indexing, retrieval, reading** as stages [R16]. For diagnosis, splitting indexing into *captured versus retained/current* and adding an *exposure/selection* and *governed-use* stage is practically necessary. Let S indicate a correct, current record in storage; R the record retrieved; E the right

evidence exposed to the model; *G* appropriate use; *Y* answer correctness. An illustrative factorization, under a specific fact/episode and explicit definitions, is

$$P(Y = 1) = P(S = 1) \times P(R = 1 \mid S) \times P(E = 1 \mid R, S) \times P(G = 1 \mid E, R, S) + \varepsilon$$

where ε covers correct answers from parametric knowledge, guessing, other sources, and paths violating the simplified chain. The formula is a **diagnostic accounting device**, not an established independent-causal model. It explicitly warns against treating answer accuracy as "memory recall."

A four-way fault-isolation test is essential: **(A)** normal full system; **(B)** inject the verified gold memory record into the real store; **(C)** force-retrieve and expose that record, preserving the rest of the prompt; **(D)** give the model only the exact gold evidence with a clean current instruction. If A fails and B succeeds, storage/writing is implicated; if B fails and C succeeds, retrieval/selection is implicated; if C fails but D succeeds, exposed-context interference/format may be causal; if D fails, answer interpretation/generation can fail even on clean evidence. Additional probes must separately test staleness, revocation, privacy, and competing timestamps. Such interventions should be instrumented; merely observing a correct answer under D is not proof that no other fault exists in A.

4.2 Benchmarks that get beyond a single "remembered?" score

LoCoMo (ACL 2024) consists of **10** human-edited, largely simulated very-long conversations, averaging **588 turns**, **27.2 sessions**, and **16,618 tokens** per conversation; its text QA and long-range temporal tasks reveal difficulties with dialogue histories, but its small number of underlying stories and multimodal components restrict generalization to natural text-only personal memory [R15].

LongMemEval (ICLR 2025) uses **500 questions** spanning information extraction, cross-session and temporal reasoning, updating, and abstention. The standard **S** history contains approximately **115K tokens**; an **M** setting uses **500 sessions and roughly 1.5M tokens** [R16]. With GPT-4o, its inspected short gold-evidence ("oracle") versus full-S comparison was **0.870 versus 0.606** QA accuracy (30.3% relative drop), despite both using the same answer questions. This contrast shows that *exposure to the full history can impair reading*; it is **not** a test of true persistence loss because the offline model directly receives history. In the M memory-retrieval ablation, augmenting each round's indexing key with extracted user facts changed **Recall@5 from 0.582 to 0.644** and GPT-4o **Top-5 answer accuracy from 0.615 to 0.657**. This is direct evidence that indexing choices can influence retrieval and downstream answers, albeit within an authored benchmark and an LLM-graded QA protocol. The paper reports grader agreement with human experts above 97%, but judge dependence and constructed histories remain limitations.

MemoryAgentBench (ICLR 2026) evaluates incremental memory retrieval, test-time learning, long-range understanding, and selective forgetting across heterogeneous agent architectures [R17]. Memora (ACL Findings 2026) stresses obsolete preferences and revisions; its **Forgetting-Aware Memory Accuracy** penalizes using invalidated information [R18]. These are relevant to correctness even if a model verbatim remembers the old fact. Their overall scores should not be compared without harmonizing the exact updates, answer definitions, and memory stores.

TimlMemBench (EMNLP Findings 2026) uses **208 reviewed samples, 1,566 sessions, 1,715 questions, and 1,821 governance/activation cases** [R19]. It reports a **canonical-memory-pool** QA control (memory systems scored **79.6–95.7%**; no-memory control **11.1%**) separately from **raw-native ingestion** of sessions. Raw-native outcomes are far more variable: **revoke 20.7–71.5%**, **supersede 18.0–80.6%**, **boundary 28.0–74.1%**, **provenance 6.2–73.4%**, across evaluated systems. These *different task views* cannot be subtracted as though they were a paired pure-storage effect. They do show why a strong QA result on an already normalized memory pool is insufficient to establish safe updating, scope control, and restraint. The paper reports human/LLM-judge cross-checks, but the governed-case ontology and bilingual sample size bound external validity.

UtilMem (EMNLP 2026; **1,717 diagnostic instances**) evaluates integrating distributed and implicitly relevant evidence, generating task-oriented summaries/plans, and resisting semantically similar distractors. Its principal negative result is that retrieved evidence does not reliably become appropriately synthesized evidence [R20]. In other words, storage/retrieval can work while answer use fails. This corroborates the pipeline distinction, not the reliability of any unnamed commercial product.

4.3 A critical deployment countertest: memory recall is not natural integration

MemUse (EMNLP 2026) includes a **randomized four-month deployment of 40 users, 1,872 sessions, and seven memory conditions** in an AI diary companion [R21]. Direct QA ranged from **19.7% to 70.1%** across conditions yet reported satisfaction did not differ materially. The authors extracted **72 user-cued memory moments** for a separate test; under the *same reconstructed context and model*, Direct QA of referenced facts scored **78.8%**, but the natural conversational response referred to only **7.9%** of the facts. Their observed association of natural integration with satisfaction is from a **small subset** and is not a causal effect on user satisfaction. Still, it decisively challenges using direct fact recall alone as the criterion for user-visible memory competence. A benchmark should measure both **availability-on-request** and **appropriate activation without explicit recall questions**, including a penalty for irrelevant or prohibited disclosures.

4.4 When can a user report identify a fault?

A single incident such as “I told the assistant last week, but it contradicted me today” establishes only the observed output and the user’s history as reported. Without store snapshots, source histories, retrieval candidates, exact exposed prompt, access settings, time/version information, and model variant, plausible causes include missing write, summarization loss, retrieval miss, wrong selection, stale data, instruction conflict, trimming, genuine in-context use failure, or simple ambiguity. Multiple controlled reproductions can support a *product-level failure rate* for a defined workflow, but still need instrumentation to identify internal mechanisms. No specific anecdote was supplied for mandatory diagnosis.

5. Q4 — Evaluation designs, metrics, and causal identification

5.1 Designs mapped to the estimand

Target effect	High-value comparison	What it can establish	Main unresolved variable
Placement	Same task and evidence, randomized gold passage slot	Causal effect of tested placement	Retrieval vs reading mechanism
Additional context	Same item/evidence, multiple lengths and distractor types	Effect of constructed context augmentation	Length vs semantic interference/positions
Retrieval-independent use	Separate exact retrieval probe; oracle evidence; masked/whitespace controls	Residual degradation despite observed retrieval capacity	Internal state during answer trial
Turn structure	FULL vs CONCAT vs SHARDED; fixed evidence and length	Effect of sharding and role/order protocol	Pure turn count vs delayed disclosure/self-text
Self-generated history	Oracle/gold vs self-conditioned prior answers	Contribution of error propagation	Hidden representation and token confounds
Persistent memory	Normal, gold-write, forced-retrieve/expose, gold-reader	Stage-specific *intervention effect*	Unlogged product architecture and interference
Freshness/governance	Update/revoke/scope/cue tests with exact chronology	Quality of *eligible use*, not just recall	Policy interpretation and grading validity
Real-world utility	Randomized deployment, blinded fact/rule annotation	Outcome in observed population/workflow	Low power, selection, external validity

5.2 Metric discipline

Answer accuracy: **Acc = (1/N) Σ_i 1[$\hat{y}_i = y_i$]** is straightforward only when a unique verified answer exists. For generative QA report rubric, evidence-based evaluation, human audit, and disagreement. **Recall@k**: fraction of all annotated relevant evidence units recovered in top *k*, with a denominator that accounts for multiple gold pieces; **NDCG@k** incorporates ranked graded relevance. Neither guarantees that the model uses retrieved facts. **Factual support precision**: **Precision(facts) = (supported atomic claims) / (checkable atomic claims)**, with coverage or completeness reported separately [R22]. **Version-valid accuracy** checks whether the answer is true *at the requested*

time. **Abstention calibration** reports selective risk *R(c)* as a function of coverage *c* (or error/unsupported-assertion rate conditional on answering), including false refusals. **Governed-use metrics** explicitly count stale-fact reuse, invalidated facts, unrequested activation, privacy/scope violations, and missing source provenance [R18,R19]. **Run-to-run reliability** in the Laban protocol is the $\mathbf{P_{90} - P_{10}}$ score spread, which must not be confused with truth probability [R11].

Inferential reporting. Use paired bootstrap confidence intervals or clustered models where variants reuse the same base question, source story, participant, or simulated conversation; do not count multiple fact questions over one story as independent people. Keep all random seeds, temperatures, sampling repetitions, run order, truncation policy, tokenization, model IDs/version dates, system prompts, tools, and context-window settings. Report absolute differences in percentage points and relative declines with denominators explicitly marked. Correct for multiple contrasts if a broad search across prompts/positions is used. Separate exploratory ablations from preregistered confirmatory claims.

5.3 A proposed minimal factorial benchmark (new design, not executed)

Use a matched corpus of verified factual questions requiring one or multiple evidence pieces, plus temporal-update and no-answer cases. Cross: **length** (e.g., 4K, 16K, 64K), **gold position** (start/middle/end), **distractor** (none/lexically similar/conflicting), **turns** (1/2/4/8), **history** (neutral/gold/self-generated), and **memory exposure** (no memory/gold store/forced retrieval/direct gold input). For answer items, construct and check semantic equivalence across turn templates; tokenize precisely. Randomize treatment order, hold final instructions stable, include a closed-book/no-evidence baseline, test across models and repeated temperatures, and preregister primary contrasts. This factorial has **3×3×3×4×3×4 = 1,296 cells per model** before repetitions, so a blocked fractional factorial and staged studies are more tractable. Retain instrumented traces to test alternative causes, and audit human rubric judgments.

The key discriminant is whether providing the *same correct evidence* by an oracle route removes the error. If it does, availability/selection becomes suspect; if not, the remaining failure lies at least partly in interpretation, answer construction, or conflicts elsewhere in the active prompt. A change in performance after an intervention identifies the intervention's effect; it does not necessarily diagnose a unique component when the intervention changes multiple things at once.

6. Q5 — What commercial-system evidence can support

Vendor documentation and internal evaluations. Google's 2024 Gemini 1.5 announcement reported approximately **99%** success on single-needle extraction up to 1M tokens [R26]. OpenAI's April 2025 GPT-4.1 announcement claimed successful one-needle retrieval up to 1M, while its more demanding internal multiple-needle **MRCR** results for GPT-4.1 were **57.2% at 128K** and

46.3% at 1M in the two-needle setting; its **Graphwalks BFS** figures were **61.7% below 128K** versus **19.0% above 128K** [R23]. The different-length test groups and difficulty distributions are not a randomized length-only estimate, but the *within-vendor juxtaposition* cautions against equating NIAH success with reliable multi-step use. Vendor figures document published testing claims and model identifiers, **not independent product reliability**.

OpenAI's **June 4, 2026** product description says ChatGPT's memory evolved from explicit saved memories through background "dreaming" synthesis to a newer summarized-memory architecture, and describes internal evaluations for carryover, preferences, and freshness [R24]. This documents that cross-session memory involves distinct background curation and relevance decisions; it does not disclose full per-answer storage/retrieval traces or establish that any particular user-reported error arose there. Settings, account entitlements, and software versions may vary. A dynamically updated help article is useful for feature definitions but unsuitable as an unversioned time-series measurement [R25].

Independent controlled product pilots. LongMemEval's researchers manually tested **97 questions** in short histories over **3–6 sessions** through web interfaces in the first two weeks of **August 2024**. For the then-tested ChatGPT/GPT-4o configuration, QA accuracy was **0.5773**, while offline GPT-4o reading of the same short histories scored **0.9184**; Coze/GPT-4o scored **0.3299** [R16]. This is evidence about those historical configurations and evaluation protocol; it cannot be relabeled "ChatGPT 2026 memory accuracy" or attributed solely to a storage failure because the offline and online systems differ in prompt handling, memory selection, and product architecture. Independently authored benchmark results can be valuable even when the team itself developed the test; this is not an exhaustive independent audit of commercial features.

User incidents and platform updates. Anecdotes can reveal failure hypotheses, particularly stale preferences or an answer that changes after restarting a chat, but they rarely reveal the precise active context or memory read. Hidden system instructions, retrieval, summaries, automatic trimming, release changes, permissions, and model routing are confounders. Product-by-product rankings and extrapolation to a current service from earlier API model scores are outside scope. The review found no inspected public, randomized, instrumented, cross-session comparison sufficient to estimate a contemporary causal storage-vs-retrieval-vs-reading share of errors across major commercial assistants. This is a **bounded evidence gap**, not a claim that such data do not exist.

7. Integrated answers to Q0–Q5

Q0 — Overall conditions and separation. Repeated controlled tasks reveal sensitivity to gold-evidence placement, distractor density, conflicting content, semantic retrieval difficulty, context length even under retrieval-capable probes, incremental disclosure, and dependence on earlier model outputs

[R1,R6,R8,R11,R14]. Cross-session persistence is a *different intervention surface*; outputs alone underidentify its cause [R16,R19-R21]. The relationship is conditional and can be nonmonotonic.

Q1 — Effective context. Strong for positional effects and task-dependent effective limits, with useful counterexamples at long lengths. High task-to-task heterogeneity; a capacity advertisement or vanilla needle result is insufficient. Attention-based causal explanations remain partially unisolated [R1,R3,R6,R8,R10].

Q2 — Conversation dynamics. Evidence is substantial for the *multi-turn sharding/self-conditioning treatment* at well-below-window lengths, and for temporal-state error propagation. The incremental causal effect of **turn count alone, strictly holding total tokens and semantics fixed**, is incompletely identified. Oracle/gold repetition can improve accuracy before it worsens [R11,R14].

Q3 — Cross-session memory. LongMemEval provides directly measurable retrieval and reading effects; TimlMemBench, Memora, UtilMem, and MemUse show that currency, relevance activation, controlled withholding, and natural use must be tested separately. A failure in one later reply alone cannot classify the stage [R16,R18-R21].

Q4 — Designs and attribution. Paired position permutations, token-/content-matched turn sharding, gold versus self-conditioned history, forced-retrieval/gold-write interventions, and transparent evidence-based scoring supply the best available tests. No one metric stands for retrieval, reasoning, factuality, or correct memory governance; reported aggregate accuracy and score spread must be operationally defined.

Q5 — Products. Vendor materials establish documented architectures and tested claims; independent evaluations establish behavior under particular versions, interfaces, and datasets; anecdotes establish experienced output discrepancies, not internal mechanisms. Product version drift and hidden context/retrieval prevent general causal inference without traces [R16,R23-R26].

8. Implications, priorities, and limitations

Practical evaluation priority. Before claiming a “memory bug,” capture the last known correct source fact, its timestamp/scope, the memory store or summary snapshot when accessible, retrieval/exposure traces, the final input actually shown to the reader, and repeated outputs. Report whether gold evidence in the current prompt fixes the issue. For long-context QA, test at least beginning/middle/end positions and both lexical and semantically indirect evidence; for multi-turn tasks compare full, concatenated, sharded, and oracle-history variants. For deployed memory, score *appropriate activation and restraint*, not just exact recall [R1,R6,R11,R16,R19-R21].

Limits of this review. This was a targeted English-language web discovery of public primary sources and documentation, not an exhaustive corpus, preregistered search, or meta-analysis. Some sources were examined at full-paper level and others at abstract/proceedings level; no new model inference, reverse-engineering of commercial storage, external replication, or direct inspection of user accounts was conducted. The original TACL paper, ICLR paper, and several 2025–26 PDF articles were inspected; the evidence matrix in the companion identifies depth per source. Model generations, tasks, languages, annotators, graders, and sampling protocols differ. The **39%**, **30.3%**, **22.0 points**, **79.6–95.7%**, and **71-point gap** reported here are not commensurable pooled effects. Some apparent failure mechanisms are author explanations rather than identified causes. Search discovery can miss unpublished null results or restricted product telemetry. Consequently, these findings support **conditional, mechanism-aware evaluation**, not a law that long conversations necessarily make assistants less factual.

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